Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building reusable systems, data-backed applications, and CI/CD-tested workflows with strong grounding in object-oriented design, problem solving, and product reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Designed and optimized multi-tenant relational schemas across **Azure SQL, PostgreSQL, and SQL Server** for **500+ users** and **10K+ structured records**, cutting dashboard latency by **71%** (**1.2s to 350ms**).
- Built reusable backend workflows with **FastAPI, Supabase, and cloud storage**, processing **5K+ monthly jobs** with **99%+** reliability through validation checks, release automation, and failure isolation.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Engineered a production grading platform backend handling **10K+ requests/day** using **Flask, MySQL, AWS RDS, and REST APIs**, with concurrent processing for **1K+** uploaded files and stable peak-load behavior.
- Implemented structured grading, rubric, and usage-tracking workflows for **150+ instructors**, improving turnaround time by **60%** while increasing reliability, visibility, and maintainability of the platform.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform software with **microservices**, reusable OOP modules, and **20+** unit/integration test suites using **Jenkins, PyTest, and Mockito** across Windows, Linux, and macOS.
- Optimized APIs and ETL pipelines with **AWS Glue, S3, and design patterns**, improving response times by **30%**, increasing pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **microservice-based resource platform** with **MERN**, WebSockets, and Azure SQL, synchronizing **2M+ records** across locations and supporting **8** business workflows for allocation, billing, and notifications.
- Automated build, test, and deployment workflows with **Jenkins** and Docker, improving release validation to **95%+**, reducing resource wastage by **42%**, and increasing lead conversions by **15%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Flask** APIs and real-time Firebase workflows for a cross-platform service platform handling **50K+ records**, improving booking confirmation speed by **40%** and reducing manual refund handling by **70%**.
- Designed pricing, payment, and analytics abstractions with **Databricks, Spark, and webhooks**, sustaining sub-**200ms** API latency and supporting hourly operational reporting.

Skills

Languages: Java, Python, JavaScript/TypeScript, SQL, C++
Engineering: Object-Oriented Design, Data Structures, Algorithms, Problem Solving, Reusable Systems
Backend & Data: FastAPI, Flask, Node.js, REST APIs, PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB
Quality & Delivery: PyTest, Mockito, Jenkins, GitHub Actions, CI/CD, Test Automation, Debugging, Documentation

Projects

FinNexus Live | *Next.js, Kubernetes, MongoDB, Plaid*
- Built a fintech platform supporting **50K+ users** and **1M+ transactions**, implementing secure transaction workflows, real-time updates, and CI/CD-driven release automation in a regulated-style domain.

Chat2SQL GitHub | *Java, TypeScript, Docker, PostgreSQL*
- Built a SQL development workspace supporting **10+** databases, enabling schema exploration, SQL generation, and structured query execution across a Java and TypeScript codebase.

CPU-scheduling Simulator Live | *C++, React.js, EJ-2*
- Implemented FCFS, SJF, SRT, RR, priority, and MLQ algorithms with dynamic visualization, reducing debugging time by **20%** and improving average turnaround time by up to **25%** across **1K+** simulations.